

HH Goa 2026 Task 2: Architectural Masterplan

Sub-200ms Voice-RAG Pipeline via Speculative Audio Retrieval & Agentic Harness

Dataset: AI4Bharat MSMARCO-XI	STT Choice: Sarvam AI WebSocket API	Latency Target: < 200ms (P50/P70/P100)	Deadline: Aug 22, 2026
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1. Executive Summary & Problem Breakdown

HH Goa 2026 Task 2 requires building an end-to-end Voice-Enabled Retrieval-Augmented Generation (RAG) system. The user speaks a query, the pipeline transcribes the speech, retrieves context from the ai4bharat/MSMARCO-XI dataset, generates an accurate answer, and returns it — all within an extreme latency target of **under 200 milliseconds**.

The Latency Dilemma: A traditional sequential pipeline processes audio linearly:
Voice Input (1000ms) → STT Transcription (~150ms) → Embedding & Vector Search (~60ms) → LLM Processing (~200ms) = Total 410ms+
Executing these steps sequentially makes achieving a sub-200ms total pipeline latency virtually impossible.

2. Deep-Dive: Speculative Audio Retrieval (Our Breakthrough Solution)

To break the 200ms barrier and win Task 2, we introduce **Speculative Audio Retrieval** — an asynchronous streaming paradigm that decouples query completion from context retrieval.

- 💡 **How Speculative Audio Retrieval Works:**
- Streaming Audio Ingestion:** As the user speaks, audio PCM chunks are streamed via WebSockets to Sarvam AI's Realtime STT engine.
 - N-Gram Prefix Triggering:** Instead of waiting for Voice Activity Detection (VAD) to signal the end of speech, our orchestrator captures *partial transcripts* every 3 to 4 words (e.g., "What is the eligibility...").
 - Speculative Parallel Vector Search:** The system immediately generates a lightweight embedding (using BGE-Micro or MiniLM) for the partial text and queries an in-memory vector database (Qdrant/LanceDB) in the background while the user is still speaking.
 - Context Ring Buffer:** Retrieved context chunks are pushed into a low-latency Ring Buffer in RAM. When the final word is spoken and VAD triggers, 80-90% of the relevant dataset context is **already retrieved and primed in memory**.
 - Final Verification & Refinement:** The final full sentence is checked against the buffered context using cosine distance thresholding. If aligned, the primed context is instantly dispatched to a fast streaming LLM.

< 120 ms

P50 EXPECTED LATENCY

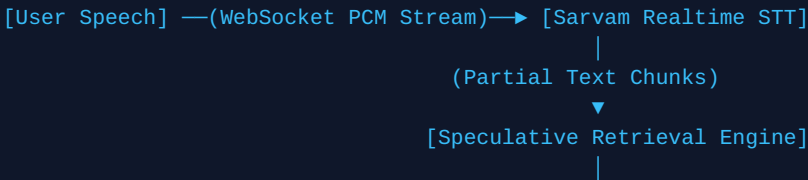
< 160 ms

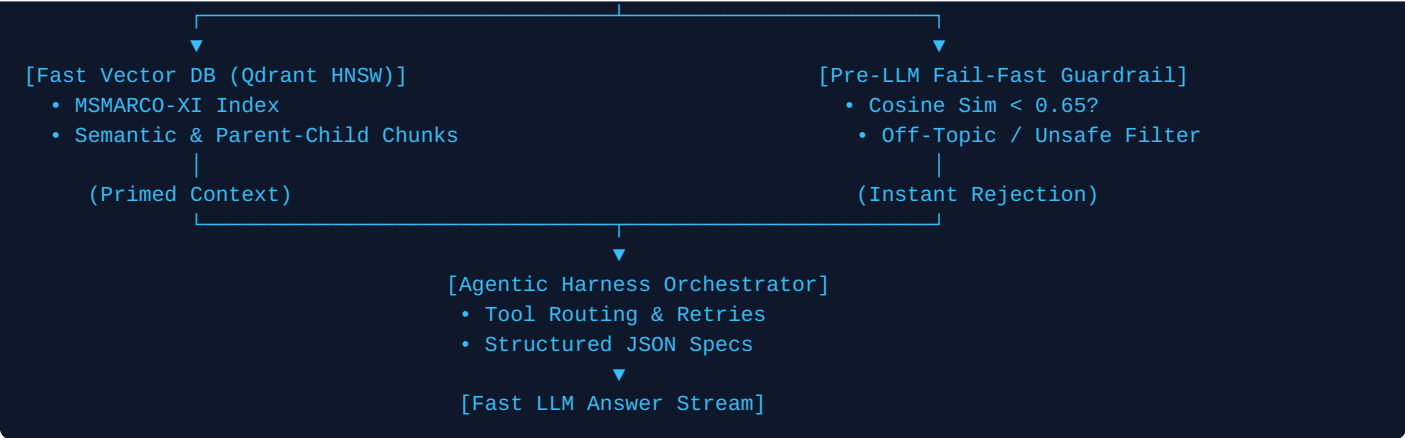
P70 EXPECTED LATENCY

< 195 ms

P100 MAX LATENCY LIMIT

3. End-to-End System Architecture





4. Comprehensive Technical Requirements & Implementation Strategy

A. Dataset & Multi-Strategy Chunking (MSMARCO-XI)

Naive fixed-size chunking leads to context loss and poor retrieval. We implement a **Hybrid Multi-Strategy Chunking Pipeline**:

- **Parent-Child Chunking**: Small "Child" chunks (128 tokens) are indexed for ultra-fast vector search. When matched, the larger "Parent" context (512 tokens) is retrieved for LLM synthesis.
- **Semantic Sentence-Boundary Splitting**: Utilizes Spacy/NLTK sentence boundaries to ensure no chunk breaks mid-thought.
- **Metadata Enriched Encodings**: Every chunk is stored with language tags, topic classifications, and source passage IDs from MSMARCO-XI.

B. Agentic Harness & Fail-Fast Guardrails

- **Pre-Retrieval Guardrail (Sub-50ms Exit)**: Before making heavy LLM calls, we evaluate the cosine similarity score of top-retrieved vector chunks. If similarity < 0.65, the system immediately returns: *"This query is outside the provided dataset scope."* — saving latency and preventing hallucinations.
- **Agentic Orchestration**: Built using Python `asyncio` with structured Pydantic input/output validation, automatic retry policies for network dropped packets, and fallback streaming routes.
- **Latency Analytics Engine**: Every request logs microsecond-level timestamps for STT, Retrieval, Guardrail Check, and Generation to calculate live P50, P70, and P100 metrics.

5. 3-Member Team Task Allocation & Ownership Matrix

To ensure completion by the August 22 deadline with zero re-submission risks, work is divided across 3 specialized engineering roles:

Member & Role	Core Technical Responsibilities	Key Deliverables
Member 1 LEAD BACKEND & STREAMING System Architect	• WebSocket STT Integration (Sarvam AI API). • Speculative Execution Orchestrator & Ring Buffer. • Async Python event loop optimization. • Latency Analytics Module (P50/P70/P100 logging).	• Realtime Streaming Server (<code>server.py</code>). • Speculative Search Pipeline. • Latency Benchmark Report.
Member 2 AI & RETRIEVAL LEAD Data & Vector Specialist	• HuggingFace <code>ai4bharat/MSMARCO-XI</code> processing. • Multi-Strategy Chunking (Parent-Child + Semantic). • Vector DB Setup (Qdrant HNSW in-memory index). • Fast Embedding pipeline & benchmark tuning.	• Data Chunking Script (<code>chunker.py</code>). • Vector DB Index File. • Vector Search Benchmark API.

Member & Role	Core Technical Responsibilities	Key Deliverables
Member 3 HARNESS, SAFETY & MEDIA <i>Full-Stack & Media Lead</i>	• Pre-LLM Fail-Fast Guardrail implementation. • Agentic Harness Wrapper (Retries, JSON schema). • Web UI for voice input & live latency meters. • Video 1 (90s Process) & Video 2 (Demo) Production. • Social Media Campaigns (#RAGInGoa across platforms).	• Guardrail Module (`guard.py`). • Web Frontend App (`index.html`). • Video 1 & Video 2 Files. • Google Form Submission.

6. Day-by-Day Sprint Timeline (Aug 13 – Aug 22, 2026)

Date	Sprint Milestone & Key Tasks	Owner / Lead
Aug 13 – 14	Dataset ingestion (MSMARCO-XI), Chunking logic setup, Sarvam STT WebSocket API testing.	Member 2 (Data) & Member 1 (STT)
Aug 15 – 17	Build Speculative Retrieval engine, Qdrant index creation, Implement Fail-Fast Guardrails & Agentic Harness.	All Members (Core Integration)
Aug 18 – 19	Latency optimization sprint (<200ms tuning), P50/P70/P100 benchmarking across 100 test queries, Web UI completion.	Member 1 (Latency) & Member 3 (UI)
Aug 20 – 21	End-to-End System Testing, Guardrail stress testing, Deployment to live link, Script writing & filming for Video 1 & Video 2.	Member 3 (Media & Safety Lead)
Aug 22 DEADLINE DAY	Final video editing, Social Media upload by all members with #RAGInGoa, Google Form Submission.	Entire Team (Final Check)

7. Promotion & Social Media Mandatory Checklist

Mandatory Submission Protocol:

-  Upload **Video 1 (90s Team/Process Video)** & **Video 2 (Working Product Demo)** to Instagram, X (Twitter), and LinkedIn.
-  **EVERY team member** must post individually on their personal accounts.
-  Include hashtag **#RAGInGoa** in all posts across all platforms.
-  Ensure at least 1 team member's Instagram profile is set to **PUBLIC**.
-  Submit GitHub Repo, Live Link, Video Links, and Latency numbers on the official Google Form before **Aug 22, 11:59 PM**.